

```
1 //LAB CODE 0002
2 //JASON JOHNSON
3 //CS1B
4 //1/26/26
5
6 #include <iostream>
7 #include <iomanip>
8
9 using namespace std;
10 void getScore(double&, double&, double&, double&, double&);
11 double calcAverage(double, double, double, double, double);
12 char determineGrade(double);
13 void isValid(double, double, double, double, double);
14 void displayGrade(char grade);
15
16
17 int main() {
18     double test1{};
19     double test2{};
20     double test3{};
21     double test4{};
22     double test5{};
23     double avg{};
24     char grade{};
25
26     getScore(test1, test2, test3, test4, test5);
27
28     isValid(test1, test2, test3, test4, test5);
29
30     avg = calcAverage(test1, test2, test3, test4, test5);
31
32     grade = determineGrade(avg);
33
34
35     displayGrade(grade);
36
37
38     return 0;
39 }
40
41
42
43
44
45 void getScore(double& test1, double& test2, double& test3, double& test4, double& test5) {
46     cout << "Please enter 5 scores.\n\n";
47
48     cout << "Score 1: ";
```

```
49     cin >> test1;
50
51     cout << "\nScore 2: ";
52     cin >> test2;
53
54     cout << "\nScore 3: ";
55     cin >> test3;
56
57     cout << "\nScore 4: ";
58     cin >> test4;
59
60     cout << "\nScore 5: ";
61     cin >> test5;
62 }
63
64
65
66
67 double calcAverage(double test1, double test2, double test3, double test4, double test5) {
68     return (test1 + test2 + test3 + test4 + test5) / 5;
69 }
70
71
72
73
74 char determineGrade(double avg) {
75     if (avg >= 90) {
76         return 'A';
77     }
78
79     else if (avg < 90 && avg >= 80) {
80         return 'B';
81     }
82
83     else if (avg < 80 && avg >= 70) {
84         return 'C';
85     }
86
87     else if (avg < 70 && avg >= 60) {
88         return 'D';
89     }
90
91     else
92         return 'F';
93 }
94
95 void isValid( double test1, double test2, double test3, double test4, double test5) {
```

```
96     while (test1 < 0 || test2 < 0 || test3 < 0 || test4 < 0 || test5 < 0) {
97         cout << "\nPlease enter values larger than 0\n";
98
99         cout << "Score 1: ";
100        cin >> test1;
101
102        cout << "Score 2: ";
103        cin >> test2;
104
105        cout << "Score 3: ";
106        cin >> test3;
107
108        cout << "Score 4: ";
109        cin >> test4;
110
111        cout << "Score 5: ";
112        cin >> test5;
113    }
114 }
115 }
116
117 void displayGrade(char grade) {
118     switch (grade) {
119         case 'A': cout << "Your grade is an A\n";
120                 break;
121
122         case 'B': cout << "Your grade is an B\n";
123                 break;
124
125         case 'C': cout << "Your grade is an C\n";
126                 break;
127
128         case 'D': cout << "Your grade is an D\n";
129                 break;
130
131         case 'F': cout << "Your grade is an F\n";
132                 break;
133
134         default: "Error";
135
136     }
137 }
138
139
140 //OUTPUT:
141
142
143 /*
144
```

```
145 Please enter 5 scores.
```

```
146
```

```
147 Score 1: 90.3
```

```
148
```

```
149 Score 2: 120
```

```
150
```

```
151 Score 3: 78.3
```

```
152
```

```
153 Score 4: 77.9
```

```
154
```

```
155 Score 5: 85.9
```

```
156 Your grade is an A
```

```
157
```

```
158 C:\Users\jjohnson257\source\repos\Lab 1 CS1B\x64\Debug\Lab 1 CS1B.exe  
    (process 25860) exited with code 0 (0x0).
```



```
159 Press any key to close this window . . .
```

```
160
```

```
161
```

```
162
```

```
163 */
```

```
164
```